

- 1 (a) (i) angular speed = $2\pi/T$
 $= 2\pi/(3.2 \times 10^7)$
 $= 1.96 \times 10^{-7} \text{ rad s}^{-1}$ C1 A1 [2]
- (ii) force = $mr\omega^2$ or force = mv^2/r and $v = r\omega$ C1
 $= 6.0 \times 10^{24} \times 1.5 \times 10^{11} \times (1.96 \times 10^{-7})^2$
 $= 3.46 \times 10^{22} \text{ N}$ A1 [2]
- (b) (i) gravitation/gravity/gravitational field (strength) B1 [1]
- (ii) $F = GMm/x^2$ or $GM = r^3\omega^2$ C1
 $3.46 \times 10^{22} = (6.67 \times 10^{-11} \times M \times 6.0 \times 10^{24})/(1.5 \times 10^{11})^2$ C1
 $M = 1.95 \times 10^{30} \text{ kg}$ A1 [3]
- 2 (a) obeys the law $pV/T = \text{constant}$ or any two named gas laws M1